

Diffusion Models — Practice

Latents, text conditioning, and classifier-free guidance

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ES 667 · Deep Learning · IIT Gandhinagar

**Turn the theory into a working
system**

Recall the minimal theory

L21 gave us one denoiser and one reverse update:

$$(z_t, t) \xrightarrow{\varepsilon_\theta} \hat{\varepsilon} \xrightarrow{\text{reverse update}} z_{t-1}$$

Everything else in a modern text-to-image system is an **engineering decision** around this loop:

representation (what space?) · **architecture** (which denoiser?) · **condition** (what steers it?) · **sampling** (how fast?)

The question for this lecture

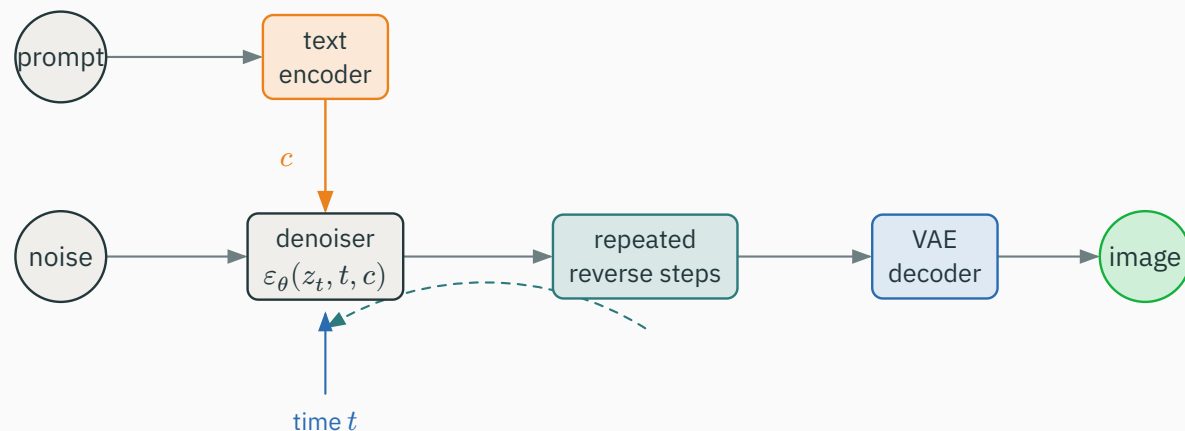
Not “how do we memorize Stable Diffusion?” but:

where does each input — noise level, text, image, mask, control — enter the denoising computation?

one worked mechanism: **classifier-free guidance** · one diagram: **latent diffusion end-to-end**

The system map **v**

Every input joins the same denoising loop at a specific place:



noise $\rightarrow$ denoiser (conditioned on time + text) $\rightarrow$ repeated reverse steps $\rightarrow$ decode $\rightarrow$ image

Why not diffuse raw pixels?

A single $512 \times 512 \times 3$ RGB image is a very large tensor:

$$512 \times 512 \times 3 = \text{786,432} \text{ values}$$

The reverse process calls the denoiser **dozens to hundreds of times**. Denoising a 786k-value tensor at every step is expensive.

can we run the **same** diffusion loop on a much **smaller** tensor?

The latent-diffusion idea

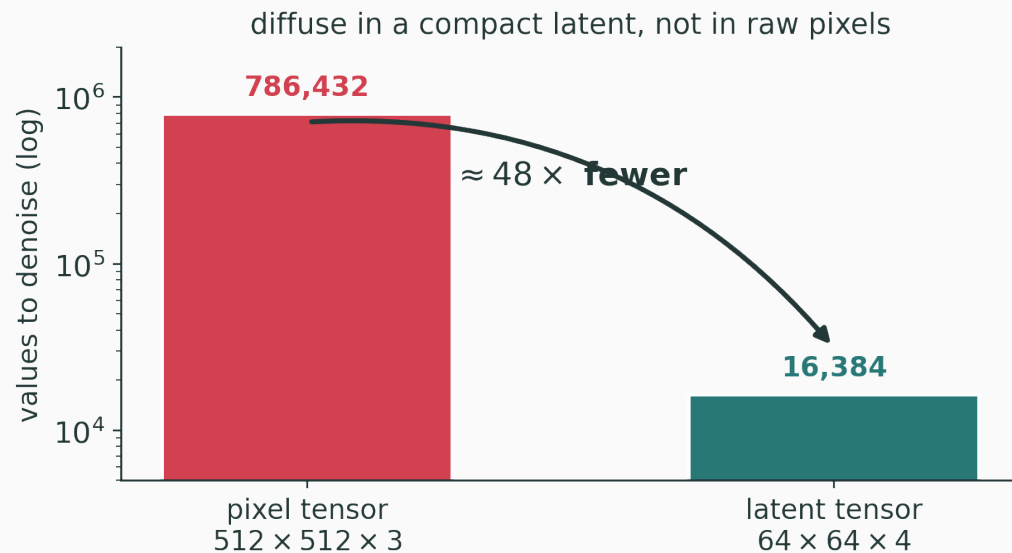
Compress the image into a compact latent with a VAE (L19), diffuse **there**, decode at the end:

$$x \xrightarrow{E} z \quad \text{diffuse in } z\text{-space,} \quad z_0 \xrightarrow{D} \hat{x}$$

Reuse the VAE's **continuous, compact** representation. The denoiser never touches raw pixels during the loop — only at the encode/decode ends.

Why latents pay off **v**

The same 512×512 image, encoded to a $64 \times 64 \times 4$ latent:



$786,432 \rightarrow 16,384$ values — about $48 \times$ fewer to denoise at **every** reverse step

What the autoencoder must preserve D

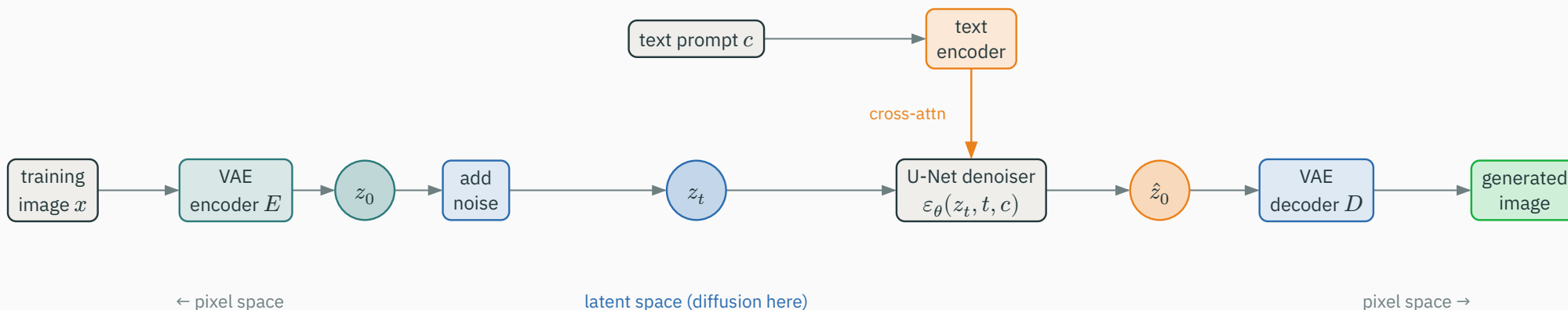
The latent is **not** just a JPEG compressor:

- it should retain **perceptual content** while cutting spatial cost;
- its **decoder** defines the ceiling on what the diffusion model can ultimately render.

If the VAE cannot decode a detail, **no amount** of diffusion in latent space can put it back. The decoder is the renderer.

Latent diffusion, end to end **v**

The full picture — pixel space at the ends, diffusion in the middle, text entering the U-Net:



encode $\rightarrow$ noise $\rightarrow$ denoise (conditioned on t and c) $\rightarrow$ predicted clean latent $\hat{z}_0 \rightarrow$ decode

The training loss stays familiar

Only the **space** and the **conditioning input** change — the objective is L21's, unchanged:

$$\mathcal{L} = \mathbb{E}_{z_0, t, \varepsilon, c} \left[\|\varepsilon - \varepsilon_\theta(z_t, t, c)\|_2^2 \right]$$

predict the noise added to the **latent** z_t , now also given the condition c — same regression as before

Freeze or train the autoencoder?

A common latent-diffusion recipe splits the two hard problems:

1. train (or obtain) an image autoencoder;
2. **freeze** it;
3. train the denoiser in the frozen latent space;
4. decode **only** the final samples.

separates **representation compression** from the expensive **generative** training

Latent scaling matters D

The noise schedule assumes a predictable latent **magnitude**:

- latents are normalized / rescaled so that data latents and Gaussian noise have **compatible** scales;
- a fixed scaling factor makes z_0 and $\varepsilon \sim \mathcal{N}(0, I)$ numerically comparable.

The broader point: **interfaces between learned modules need numerical contracts** — the encoder's output distribution must match what the diffusion process expects.

The denoiser architecture

What structure does a denoiser need?

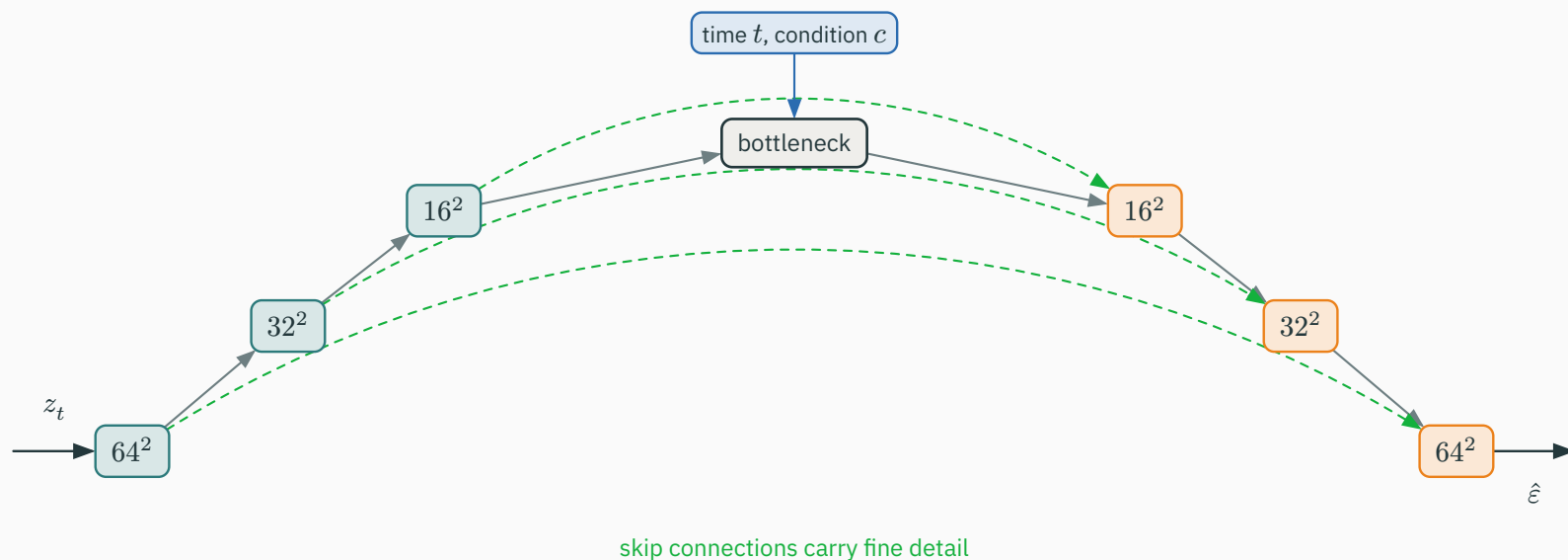
To clean a noisy latent well, the network needs four things:

- **local detail** at many resolutions;
- a **global view** of the composition;
- a **route** for conditioning information;
- knowledge of the **current noise level** t .

the U-Net (and its Transformer successors) is built to provide exactly these

The U-Net at a glance v

A symmetric encoder-decoder with skip connections:



downsample for global context $\rightarrow$ bottleneck $\rightarrow$ upsample for detail, with skips bridging matched resolutions

Why the U shape helps

The two halves solve complementary problems:

Down — pooling grows the receptive field, so the bottleneck sees **global structure** (layout, objects).

Up — skip connections re-inject **fine spatial detail** the bottleneck blurred away.

global reasoning **and** local detail — neither alone reconstructs a clean latent

The resolution hierarchy

The denoiser walks **down** then back **up** a resolution ladder:

$$64^2 \rightarrow 32^2 \rightarrow 16^2 \rightarrow \text{bottleneck} \rightarrow 16^2 \rightarrow 32^2 \rightarrow 64^2$$

- high resolution keeps **fine texture**; low resolution supplies **global context**;
- each up-step **concatenates** the matching encoder feature before convolving.

the same ladder a CNN backbone uses — here run **symmetrically** for reconstruction

Time embedding

The same noisy latent must be denoised **differently** early and late in the process:

- turn the integer timestep t into a learned vector e_t (sinusoidal $\rightarrow$ MLP);
- inject e_t into every residual block.

at $t = 900$ almost everything is noise; at $t = 20$ only a little — the model must know which

Residual blocks and timestep injection

Each block mixes spatial features h with the timestep embedding e_t :

$$h' = \text{ResBlock}(h, e_t) \quad (e_t \text{ scales / shifts the block's features})$$

The time embedding **modulates** the computation, so one visual feature can be handled one way at high noise and another way at low noise.

Skip connections carry high-frequency detail

A deep bottleneck weakens precise location and texture:

$$\underbrace{\text{encoder feature}}_{\text{at a resolution}} \longrightarrow \underbrace{\text{decoder input}}_{\text{same resolution}}$$

skips let **global reasoning** (bottleneck) and **local detail** (early layers) coexist in the output

Text is not pasted onto pixels

A text encoder turns the prompt into a **sequence of embeddings**:

$$\text{prompt} \xrightarrow{\text{tokenize}} \text{tokens} \xrightarrow{\text{encoder}} c = (c_1, \dots, c_L)$$

the denoiser never sees characters — it accesses c through **cross-attention** inside its blocks

Cross-attention reminder D

The attention you already know, with a twist on where Q, K, V come from:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(Q \frac{K^\top}{\sqrt{d}}\right) V$$

queries Q come from the latent/image features; keys K and values V come from the text tokens

Cross-attention shapes

Let the latent have N spatial positions and the prompt have L tokens:

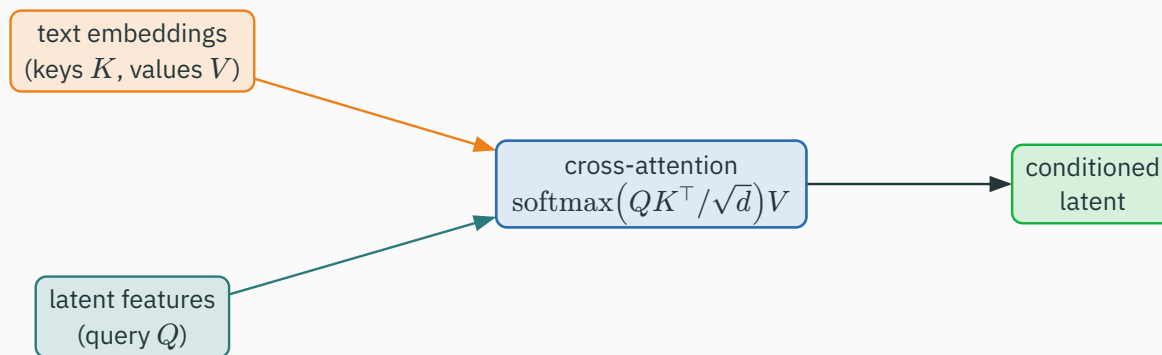
$$Q \in \mathbb{R}^{N \times d}, \quad K, V \in \mathbb{R}^{L \times d}$$

$$QK^{\top} \in \mathbb{R}^{N \times L} \quad (\text{every image position scores every prompt token})$$

the $N \times L$ attention matrix routes text information to **each spatial location**

Cross-attention as conditioning **v**

The whole conditioning mechanism in one block:



at every location the denoiser asks: **which prompt tokens matter for cleaning up this region now?**

Interactive: text conditioning I

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/text-diffusion

Type a prompt, watch it tokenize into text embeddings, then see cross-attention route each token to the image regions that attend to it during denoising.

Cross-attention interpretation D

For “a red bicycle beside a tree”:

- a location on the bicycle can attend more to **bicycle** and **red**;
- background locations may attend more to **tree**.

This is an **intuition for information flow**, not a proof of exact visual grounding — heads are learned and often diffuse.

Text encoder choices D

The denoiser reuses representation learning from L14–L18:

- a **pretrained** text encoder (e.g. CLIP text tower) maps tokens to **contextual** embeddings;

$$\text{prompt} \rightarrow \text{tokens} \rightarrow (c_1, \dots, c_L)$$

stronger text encoders = better prompt understanding — the conditioning is only as good as c

Practical condition types v

The same cross-attention / concatenation machinery accepts many conditions:

Condition	Example use
text	describe desired semantics / style
class label	class-conditional generation
image / low-res image	image-to-image, super-resolution
mask	inpainting
pose / edges / depth	layout and structure control

Architecture landscape — one slide

U-Net — the intuitive teaching architecture; convolutions + cross-attention at each resolution.

Diffusion Transformer (DiT) — replaces much of the conv backbone with **token** processing over latent patches.

architecture changes; the **noise-prediction objective** is the stable idea beneath both

**Make a condition matter:
classifier-free guidance**

The conditioning problem

A conditional model can produce **plausible** images that only **weakly** follow the prompt:

- training maximizes likelihood, not prompt adherence;
- we want **stronger** adherence — **without** a separate external classifier.

how do we amplify the prompt's influence using only the denoiser we already trained?

Train with occasionally dropped conditions

During training, sometimes replace c with an **empty** condition $\emptyset$:

$$\varepsilon_{\theta}(z_t, t, c) \quad \text{and} \quad \varepsilon_{\theta}(z_t, t, \emptyset)$$

One network learns both a **conditional** and an **unconditional** denoiser — the empty condition is a real, learned input (typically dropped 10% of the time).

CFG training procedure

For each training example:

1. choose the condition c ;
2. with some probability, **replace** c by the empty condition $\emptyset$;
3. sample a timestep t and noise ε ;
4. train $\varepsilon_{\theta}(z_t, t, c)$ to predict ε .

no architecture change — just occasionally blank the condition

Condition dropout is not prompt dropout D

A subtle but important distinction:

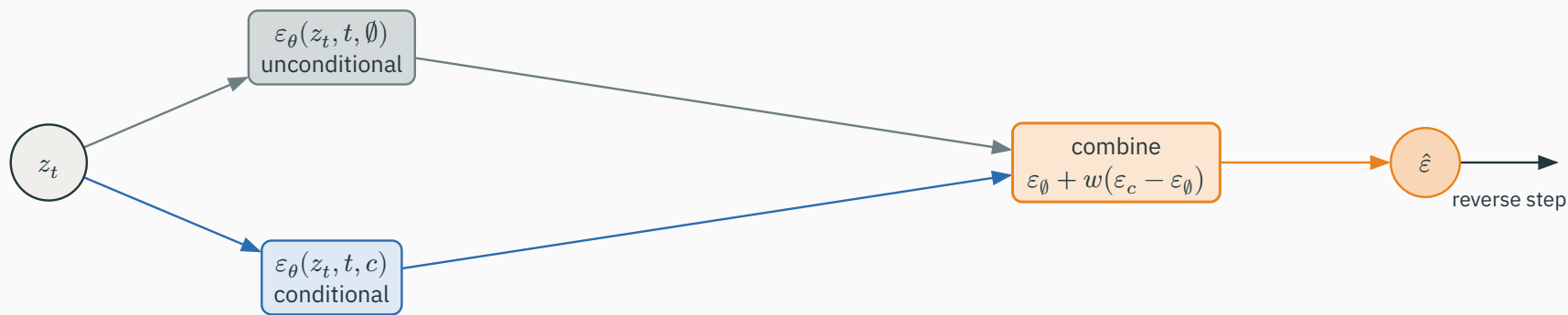
Not this — drop a few **words** from the prompt. That is still a (weaker) conditional model.

This — drop the **entire condition representation** $c \rightarrow \emptyset$, giving a genuine **unconditional** estimate.

the empty prompt must be a **meaningful learned condition** to compare against later

Classifier-free guidance **v**

At sampling time, run **both** predictions and extrapolate:



two forward passes per step, **one** network

$$\hat{\varepsilon}_{\text{guided}} = \varepsilon_\theta(z_t, t, \emptyset) + w[\varepsilon_\theta(z_t, t, c) - \varepsilon_\theta(z_t, t, \emptyset)]$$

Read the formula in words **D**

$$\hat{\varepsilon} = \varepsilon_{\emptyset} + w(\varepsilon_c - \varepsilon_{\emptyset})$$

- start from what the model would create **unconditionally** ($\varepsilon_{\emptyset}$);
- move **further** in the direction the prompt **changes** the prediction, scaled by w .

$w = 0$: ignore the prompt · $w = 1$: plain conditional · $w > 1$: **amplified** prompt influence

Board example D

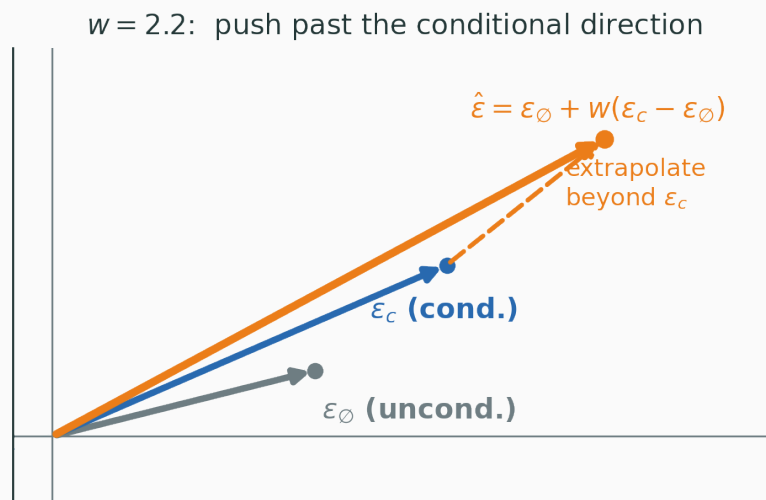
Take a scalar toy: unconditional = 2, conditional = 1.5, guidance $w = 4$:

$$\hat{\varepsilon} = 2 + 4(1.5 - 2) = 2 + 4(-0.5) = 2 - 2 = \mathbf{0}$$

The guided prediction (0) is pushed **past** the conditional value (1.5). A larger w **amplifies** the conditional–unconditional gap — it is **not** a probability.

Guidance is vector extrapolation **v**

In two dimensions, $\hat{\varepsilon}$ extrapolates **beyond** the conditional direction:



$w = 1$: land on $\varepsilon_c \cdot w > 1$: shoot past it along the $(\varepsilon_c - \varepsilon_{\emptyset})$ direction

CFG misuse check D

A common misreading:

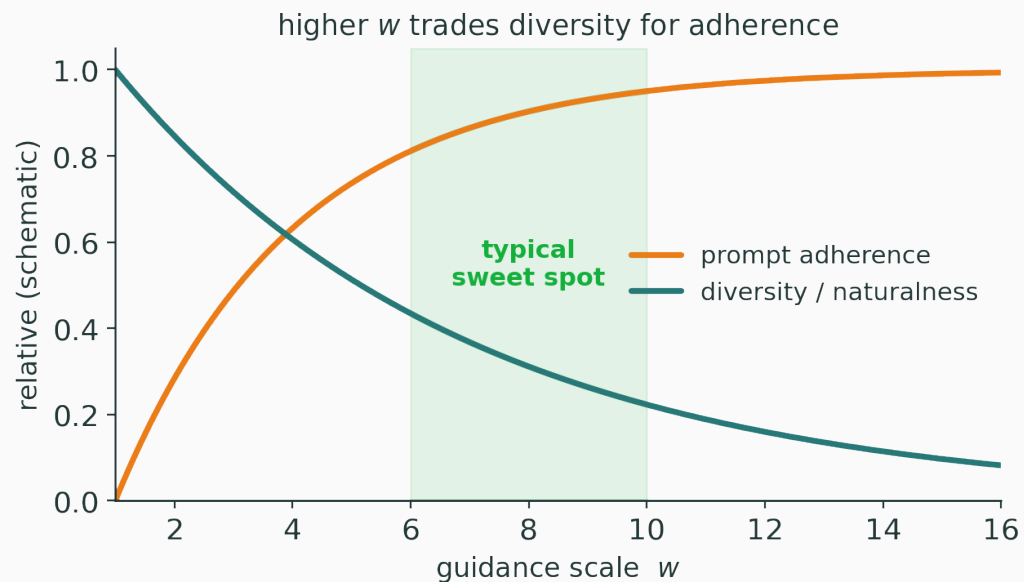
$$w \neq \Pr(\text{prompt is correct})$$

w is a **heuristic scale** on one prediction relative to another — nothing more.

Very high w over-extrapolates: samples become **less natural**, **oversaturated**, and **less diverse**.

Guidance-scale trade-off v

Sweeping w trades two things against each other:



low w : diverse, weaker adherence · high w : strong adherence, can look oversaturated / repetitive

Interactive: guidance scale **I**

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/cfg-scale-visualizer

Drag the guidance scale w and watch $\hat{\epsilon} = \epsilon_{\emptyset} + w(\epsilon_c - \epsilon_{\emptyset})$ extrapolate — see prompt adherence rise and diversity fall as w grows.

Prompt wording is distribution matching

A prompt is a **condition**, not a spell:

- different phrasing = different text embeddings c = a different steering vector;
- “steering” means **within** the learned training distribution, not conjuring the impossible.

prompt engineering = finding the c whose region of the distribution you actually want

Sampling, editing, and the engineering trade-off

Sampling is an iterative solver

The reverse update need **not** use every original training step:

- training uses a fine schedule (e.g. $T = 1000$);
- sampling can take a **shorter path** through the **same** denoiser.

faster samplers re-traverse the learned reverse field with **fewer, larger** steps

The sampler is an inference algorithm **D**

Separate two things that are easy to conflate:

$\underbrace{\text{model parameters } \theta}_{\text{trained once}}$	vs.	$\underbrace{\text{number and type of reverse updates}}_{\text{chosen at inference}}$
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The denoiser is trained **once**. Many compatible samplers can reuse its weights, but their schedule, quality, and diversity trade-offs still need to be checked.

DDPM vs accelerated sampling **v**

DDIM — few deterministic steps (e.g. 20–50)



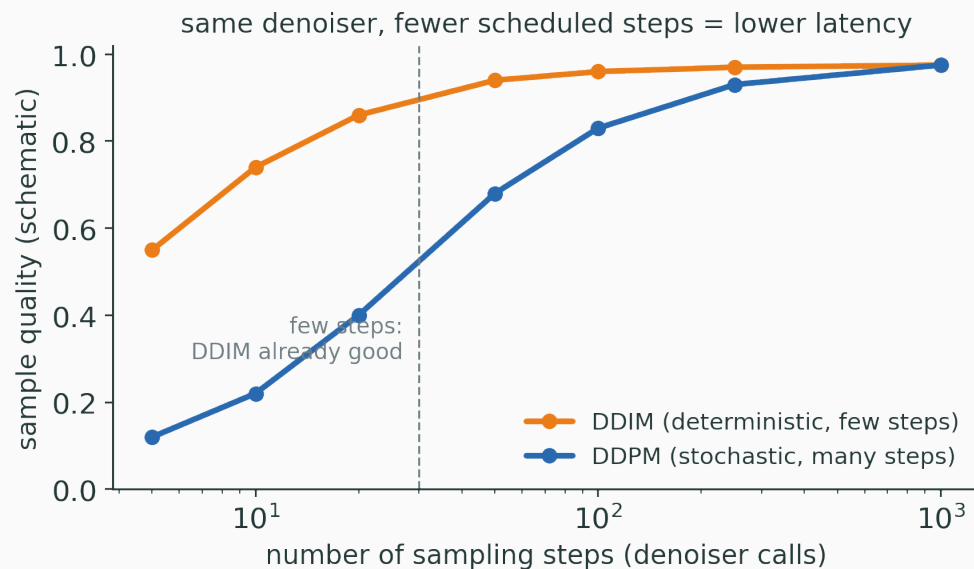
DDPM — many stochastic steps (e.g. 1000)

	Many stochastic steps	Fewer scheduled steps
benefit	faithful, simple formulation	lower latency
cost	slow generation	possible quality / diversity trade-off

DDIM is **one** example of a fast, deterministic-style sampler — we do not derive its update here

Sampling-step budget v

More steps = more denoiser calls; the returns diminish:



fewer steps: fast iteration · more steps: final rendering — **no** single count is universally optimal

Interactive: the reverse process I

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/diffusion-denoise

Step through the reverse process one update at a time, and watch how the step schedule and count change the denoising trajectory from z_T to z_0 .

Seed and reproducibility

A seed fixes the **initial latent noise** z_T :

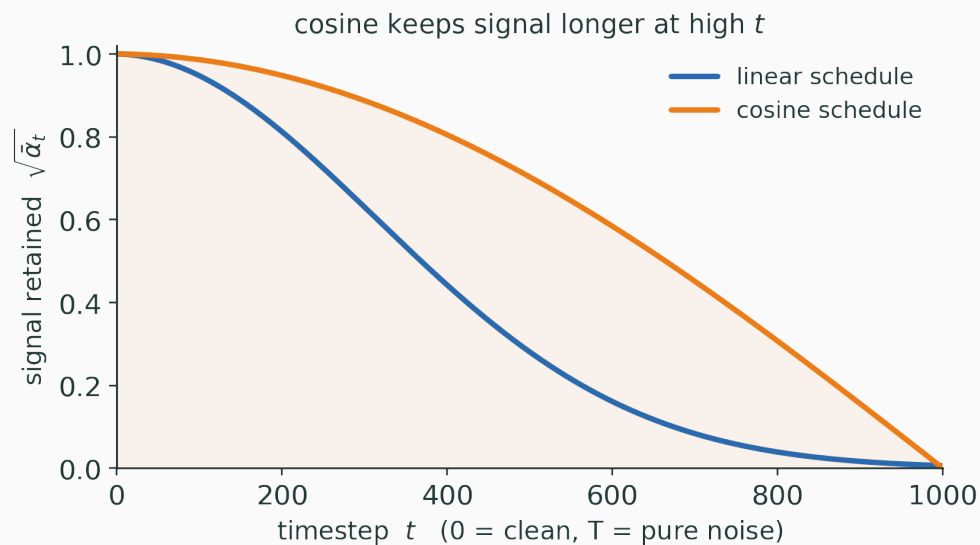
same seed $\Rightarrow$ same (or nearly same) trajectory

holding model, sampler, steps, guidance, and condition fixed.

For a scientific demonstration, **record the seed**, model version, step count, and guidance — otherwise the result is not reproducible.

Noise schedules v

The schedule sets how fast signal decays into noise — a real design knob:



cosine schedules keep **signal** $\sqrt{\alpha_t}$ longer at high t , spending more steps where it matters

Image-to-image **v**

Start from an **encoded input**, add noise only partway, then denoise under a new prompt:



more initial noise → more freedom to change the input

more initial noise $\Rightarrow$ more freedom to change the input

Inpainting

Preserve known regions, regenerate only the masked one:

- at each step, keep a **suitably noised** version of the **known** pixels outside the mask;
- let the denoiser fill the **masked** region, conditioned on visible context + prompt.

The model must keep **matching boundaries and semantics** across the visible/edited seam — editing is not a clean cut-and-paste.

Text says **what**; a structural condition says **where**:

Input condition	What it constrains
edges	outline / composition
pose	body arrangement
depth	geometry
segmentation	regions and classes

a control signal gives the denoiser an extra **layout** representation — it does not eliminate all errors

Diffusion Transformers D

Patch/token-based backbones swap the conv U-Net for Transformer blocks:

- inputs are still **noisy latent tokens** + **timestep** + **condition tokens**;
- attention replaces much of the convolutional mixing.

the architecture changes; L21's **denoising loss remains recognizable**

Fine-tuning a diffusion model

Parameter-efficient adaptation adjusts a **small** subset of weights:

- learn a new **style** or **domain** (LoRA-style, as in the LLM module);
- inherits the same adaptation trade-offs — and raises **data-consent** questions.

Do not make a first offering depend on a **live** fine-tuning exercise — use pre-computed adapters if you must demo.

Adaptation boundaries D

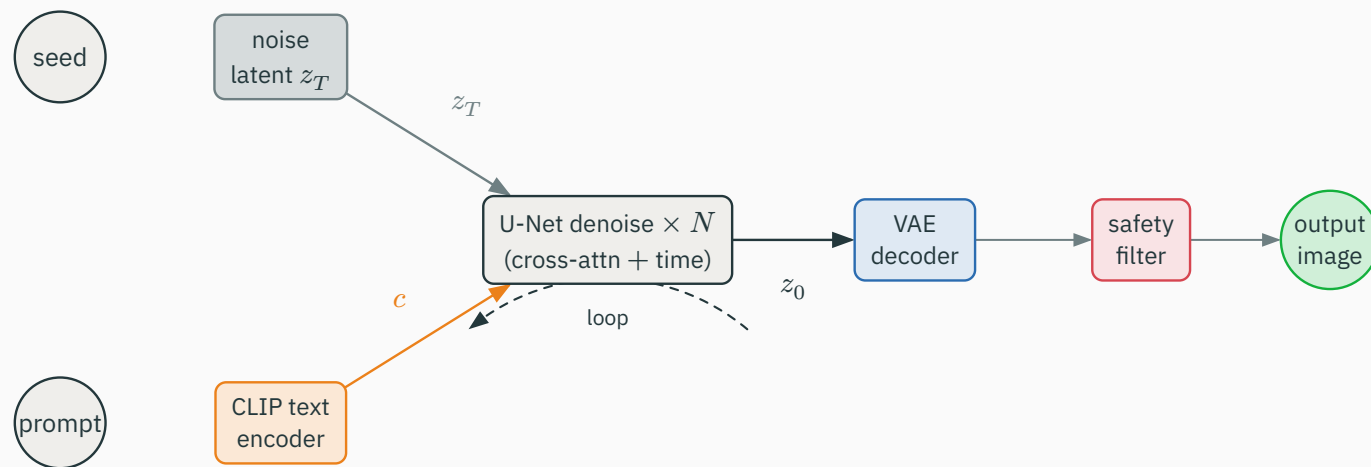
Fine-tuning can change:

- a **style** · a new **visual domain** · a **text/image association** · a **safety / refusal** behavior.

Ask: **what data licensed the adaptation**, and does the base model's capability **shift in unintended ways**?

The full Stable-Diffusion pipeline v

Putting every piece together — the system you actually run:



text + seed $\rightarrow$ conditioned denoise loop ($\times N$) $\rightarrow$ decode $\rightarrow$ safety filter $\rightarrow$ image

Where the time goes D

Not all stages cost the same:

- **text encoding** — once;
- **denoiser** — called **every** sampling step ($\times N$);
- **VAE decode** — once, at the end.

step count drives latency far more than prompt length — the loop is the bottleneck

Batch generation is not free

Generating several images together interacts with the hardware:

- better accelerator **utilization**, but **more memory**;
- latent size, denoiser architecture, and step count all interact with throughput.

this is exactly the L23 systems discussion — throughput vs latency vs memory

A small reproducible demo

Fix the seed and prompt, then change **one** variable at a time:

1. vary the **guidance scale** w ;
2. vary the **number of inference steps**;
3. change **one structural condition**.

ask students to identify which variable changed **composition**, **fidelity**, and **latency**

A reproducible classroom exercise

Prepare a grid **in advance** (seeds/settings recorded):

Intervention	Which axis it probes
fixed prompt, vary w	meaning vs. diversity
fixed seed, vary steps	fidelity vs. latency
fixed settings, vary one word	prompt sensitivity
fixed prompt, add pose / edge	composition control

A practical system checklist **D**

For a new diffusion application, **document**:

1. model and VAE **versions**
+ prompt / structural **condition**
+ **seed**

1. sampler and **step budget**
+ **guidance scale**
+ post-processing / **safety filters**

this turns an impressive image into a **reproducible experimental artifact**

Failure modes to discuss honestly

- text/image **misalignment** and unwanted objects;
- distorted **text**, **hands**, or **counting**;
- inherited **bias** and training-data **memorization**;
- high **compute and energy** cost.

fluent-looking output $\neq$ correct or safe output

Failure analysis grid

Symptom	Likely lever to inspect
ignores prompt	guidance, text condition, data mismatch
wrong composition	structural condition, seed, prompt ambiguity
weak detail	step budget, latent / decoder capacity
repeated artifacts	data / model limit, excessive guidance

a **starting point**, not a deterministic one-to-one diagnosis

Common misconceptions v

Misconception	Correction
text is painted into the image	text embeddings condition denoising via learned attention
higher guidance always improves quality	it trades adherence against naturalness / diversity
a seed is a semantic label	it selects initial noise, not a named concept
diffusion editing is exact	it is probabilistic; it may alter unintended regions

Provenance and consent D

Technical control is **not** the same as permission. Discuss separately:

- what **training data** a model may have seen;
- whether output **resembles** a living artist or a training item;
- how users should **label** generated content;
- who **bears the cost** of errors in high-stakes imagery.

capability $\neq$ accountability

Comparison with autoregressive image tokens v

Two families, two ways to factor $p(\text{image})$:

Family	Generate unit	Main inference pattern
autoregressive visual tokens	one token at a time	causal decoding
diffusion	whole latent, repeatedly	iterative denoising

both can use Transformer representations; their probability **factorizations** differ

Wrap up

The systems bridge

Every extra sampling step calls a **large** denoiser again:

N steps $\times$ one U-Net forward pass = the dominant cost

efficient inference is not an afterthought — it changes **which diffusion applications are practical**

Retrieval questions

1. Why is a VAE useful even though diffusion already provides the generative model?
2. Which tensors supply cross-attention **queries**, **keys**, and **values**?
3. What does classifier-free guidance **extrapolate**?
4. Why is a **seed** useful for a scientific demonstration?

Retrieval questions, extended

1. Why run diffusion in a **latent** space?
2. How does a U-Net combine **global context** and **detail**?
3. What does a cross-attention **query** represent in this setting?
4. Which input controls composition most directly: a **prompt** or a **pose map**?

Student takeaways

1. **Latent diffusion** cuts the cost of iterative denoising by working in a compact VAE space.
2. **Cross-attention** lets spatial denoising features retrieve relevant text information.
3. **Classifier-free guidance** trades diversity for stronger conditional adherence.
4. **Editing and control** are variations on conditioning the same reverse-denoising process.

diffusion is controllable denoising in a learned representation space

a denoiser called **many** times $\Rightarrow$ memory movement, caching, compression, and scheduling matter

Next (L23): why **efficient inference** decides what is actually usable.

Every input — noise level, text, image, mask, control
— enters the **same** denoising loop.

Next: **Efficient Inference** → memory movement, caching, and
compression.